Automated File Organizer
Algorithm and Dataset

Algorithm

Step-by-Step Procedure

1. Start the program.
2. Select the target directory.
3. Scan all files using os.listdir().
4. Check whether each item is a file.
5. Extract the file extension using os.path.splitext().
6. Compare the extension with predefined categories.
7. Create destination folders if they do not exist.
8. Move files using shutil.move().
9. Repeat until all files are organized.
10. Display a success message and end the program.

Logic / Method Used

The system uses a rule-based approach. It scans a selected directory, identifies each file by its extension, maps it to a predefined category, creates folders when necessary, and moves the files automatically.

Model / Technique Applied

• Rule-Based File Classification
• Python Automation
• File Handling
• Directory Traversal using os module
• File Movement using shutil module

Process Flow

Start → Select Directory → Scan Files → Detect Extension → Identify Category → Create Folder (if needed) → Move File → Repeat for Remaining Files → Display Success Message → End

Dataset

Dataset Description

This project does not use a machine learning dataset. The input dataset consists of files stored in a user-selected directory.

Input Dataset

Documents (.pdf, .docx, .txt, .pptx, .xlsx), Images (.jpg, .png, .jpeg), Videos (.mp4, .avi, .mkv), Audio (.mp3, .wav), Archives (.zip, .rar, .7z), and Other file types.

Output Dataset

Files are automatically organized into Documents, Images, Videos, Audio, Archives, and Others folders.